

NorthPeak Data Pipeline - Process Notes

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WARNING: Stages 1-3 only documented. Stages 4-8 NOT DOCUMENTED.

Stage 1: NP Core Extraction

Connection Details

- Host: localhost (dev) / np-core-prod.northpeak.internal (prod)
- Port: 5432
- Database: np_core
- Schema: np_core
- User: pipeline_reader (read-only)
- Connection via: SQLAlchemy + psycopg2

Tables Extracted (daily, full refresh for now - no incremental yet)

- np_core.orders (primary table - ORDER BY order_date)
- np_core.order_items
- np_core.customers
- np_core.products
- np_core.stores
- np_core.inventory
- np_core.stock_movements

Column Notes

- orders.customer_id - NULLABLE. Pre-2018 in-store transactions have no customer link.
Silver fix: map to UNKNOWN_CUSTOMER_ID = -1. Do NOT drop these rows - Finance needs them.
- orders.payment_reference - Contains card data for records before 2021-03-01.
Silver fix: SHA-256 hash this column before writing to Silver.
- inventory.quantity_on_hand - CAN BE NEGATIVE. These are DC adjustment records.
Silver fix: set is_adjustment = TRUE when quantity < 0. Do NOT treat as errors.
- customers.email - NULL for FreshMart migration batch (customer_id 401-470 approximately).
Silver fix: flag as MISSING_EMAIL, do not impute.

Known Issue - Column Naming

Pre-2018 tables (before FreshMart migration) use snake_case.

Post-FreshMart tables sometimes use camelCase or have extra prefixes.

Standardise ALL column names to snake_case in Silver.

Stage 2: GroceryDirect Extraction

Connection Details

- Type: MongoDB 6.0
- Connection string: mongodb://gd-reader:REDACTED@gd-prod.northpeak.internal:27017/grocerydirect
- Database: grocerydirect
- Collections: orders, returns, customers, delivery_slots

Key Issue - Returns Not Linked

8. 3% of return documents do NOT contain the original order_id.
They have a basket_ref field instead that SOMETIMES matches but not always.
CHECK THIS - Yemi says it is causing double-counting in revenue reports.

Key Issue - Nested JSON

order.line_items is a nested array. Explode this to row-per-item in Bronze.
Do NOT flatten at source - keep raw nested JSON in Bronze, explode in Silver.

Stage 3: SupplyLink Extraction

Connection Details

- Type: SQL Server 2019
- Host: supplylink-prod.northpeak.internal
- Port: 1433
- Database: SupplyLinkProd
- Schema: dbo
- IMPORTANT: Use pyodbc + ODBC Driver 18 for SQL Server

CRITICAL KNOWN ISSUE - SKU Silent Failures

The SupplyLink-to-EPOS handoff is failing silently for approximately 14% of SKUs.
Stock is received at DC and scanned into SupplyLink but never appears in NP Core inventory.
Marcus Webb is aware. Root cause investigation is SEPARATE ticket - NP-DATA-847.
DO NOT fix this in the pipeline - just flag it. The is_epos_synced field is your clue.
CHECK THIS with Marcus before touching anything.

CRITICAL KNOWN ISSUE - Timezone

event_timestamp in SupplyLink alternates between UTC and BST with no consistent flag.

Pattern: UTC in winter months (Oct-Mar), BST in summer months (Apr-Sep) - but NOT reliable.

Silver fix: assume UTC for all Oct-Mar records, add 1hr for Apr-Sep. Document assumption.

This needs a proper fix in a future sprint.

Column Shorthand (not self-explanatory)

- MVT_TYPE: movement type code. See glossary in company_brief.md
 - RCT = receipt from supplier
 - TRF = inter-DC transfer
 - ADJ = inventory adjustment (can produce negative quantities)
 - WOF = write-off
 - RTN = customer return processed at DC
 - IS_CONF: is_confirmed - whether movement confirmed by warehouse manager
 - DC_REF: DC internal reference, NOT the supplier delivery reference
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Stage 4 onwards: NOT DOCUMENTED

I ran out of time before leaving. The following are NOT documented:

- Stage 4: RewardsCo API extraction (pagination, rate limiting)
- Stage 5: NP Financial extraction (FCA boundary, read replica setup)
- Stage 6: Bronze to Silver transformation logic
- Stage 7: Silver to Gold / dbt setup
- Stage 8: Airflow DAG configuration

What I Know That Is Not Written Down

- RewardsCo API key is in LastPass under "NorthPeak Integrations" - ask Aisha
- The FCA read replica requires VPN + certificate - IT ticket NP-IT-2341 has details
- The Airflow Docker setup is on my local machine - I'll send you the docker-compose.yml

Open Questions (you need to answer these)

1. How do we handle late-arriving GroceryDirect returns? No SLA agreed with the business.
2. What is the correct grain for FactSales? Order-level or order-item-level? Yemi has opinions.
3. RewardsCo API - do we need historical backfill or just forward from go-live? Ask Marcus.
4. NP Financial: which tables exactly? Legal said minimum data principle applies.

Contacts

- Aisha Okafor (your manager) - architecture decisions, tool choices
 - Marcus Webb (Head of Data) - business requirements, NP Financial scope
 - Yemi Adeyemi (Analytics Engineer) - dbt consumer requirements, FactSales grain
 - Chloe Tan (BI Developer) - Power BI requirements, what columns she needs
 - Barry Higgins (SupplyLink admin, IT) - SQL Server connection issues
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